CSA0404 – OPERATING SYSTEMS – LAB PROGRAMS

EXPERIMENT 18

Question: Construct a C program to simulate producer-consumer problem using semaphores.

Program:

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>

#define SIZE 5

int buffer[SIZE], in = 0, out = 0;
sem_t empty, full;
pthread_mutex_t mutex;

void *producer(void *arg)
{
    for (int i = 1; i <= 10; i++) {
        sem_wait(&empty);
        pthread_mutex_lock(&mutex);

        buffer[in] = i;
        printf("Produced: %d\n", i);
        in = (in + 1) % SIZE;

        pthread_mutex_unlock(&mutex);
        sem_post(&full);
    }
    return NULL;
}

void *consumer(void *arg)
{
    for (int i = 1; i <= 10; i++) {
        sem_wait(&full);
        pthread_mutex_lock(&mutex);

        int item = buffer[out];
        printf("Consumed: %d\n", item);
        out = (out + 1) % SIZE;

        pthread_mutex_unlock(&mutex);
        sem_post(&empty);
    }
    return NULL;
}

int main()
{
    pthread_t p, c;

    sem_init(&empty, 0, SIZE);
    sem_init(&full, 0, 0);
    pthread_mutex_init(&mutex, NULL);

    pthread_create(&p, NULL, producer, NULL);
    pthread_create(&c, NULL, consumer, NULL);

    pthread_join(p, NULL);
    pthread_join(c, NULL);

    sem_destroy(&empty);
    sem_destroy(&full);
    pthread_mutex_destroy(&mutex);

    return 0;
}
```

Output shown in the source document:

```
Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
Produced: 6
Consumed: 6
Produced: 7
Consumed: 7
Produced: 8
Consumed: 8
Produced: 9
Consumed: 9
Produced: 10
Consumed: 10
```